

By now, you may be wondering what is the need for unsupervised learning with neural networks? After all, supervised learning has proved itself to be quite successful with neural networks. As mentioned earlier, in many cases, we are not able to apply deep learning techniques due to a lack of labelled data. And another more fundamental reason for the attractiveness of unsupervised neural network techniques is that they seem to be closer to the way human beings develop their knowledge and perception of the world. For instance, as children build their knowledge of the world around them, they are not explicitly provided labelled data. Also, more importantly, there appears to be a hierarchical learning method that human brains follow. It is the ability to represent abstract concepts, based on which higher concepts are built. If human beings have the ability to obtain much of their knowledge from unsupervised learning techniques, how can we employ similar techniques in artificial neural networks for deep learning? This requires us to take a brief digression from artificial neural networks to human cognition.

One question which has been debated a lot in human cognition is that of 'localist' vs 'distributed' representation. A localist representation means that one or a very few limited number of neurons are used to represent a specific concept in the brain. On the other hand, in a distributed representation approach, a concept is represented by distribution over a large number of neurons, and each neuron participates in representation of multiple concepts. In other words, while in a localist representation, it is possible to interpret the behaviour of a single neuron as corresponding to a specific concept/feature, in a distributed representation, it is not possible to attribute/map the activity of a single neuron in isolation as its behaviour is dependent on the activity of a host of other neurons. An extremist view of the localist approach is represented by what is known as 'grandmother cells'. Are there neurons that are associated with a single concept only? For instance, if 'grandmother' is a concept represented in the human brain, is there a single neuron that gets fired in recognising this concept? Well, an affirmative answer to that question will rule in favour of the localist approach.

There has been a raging debate on the presence/absence of grandmother cells. There have been certain experiments in the past which seemed to indicate the presence of the grandmother cell (<http://tandfonline.com/doi/full/10.1080/23273798.2016.1267782>). But recent research indicates a lack of sufficient evidence for grandmother cells. Earlier experiments indicated that certain specific neurons only fired when recognising a specific concept such as 'grandmother' or a popular personality such as Jennifer Aniston (<https://www.newscientist.com/article/dn7567-why-your-brain-has-a-jennifer-aniston-cell/>). It is possible now to think of alternate explanations which can explain these experimental results. One strong possibility is the presence of sparse coding of neurons in human brains. Sparse neural coding means that only a relatively small number of neurons are used to represent different features of a concept. For instance, let us consider an image which has 1000 x 1000 pixels. Of the million pixels, only some are used to encode

horizontal lines, certain others to encode vertical lines and so on. In fact, each sub-concept or sub-feature is represented by a selective, limited set of neurons. This means that out of the million pixels, only a few are activated for each image, enabling us to recognise a very large number of images. Thus, sparse coding provides an alternate explanation for the grandmother cell experiments. When you are shown the image of your grandmother, there are a few selective neurons that are encoding this concept and, hence, they are the ones that get activated. It is possible that these neurons also are involved in representing various other concepts. So the sparse coding explanation does not preclude a distributed representation for knowledge representation in human brains.

By now, you may be thinking that though the human brain is interesting, how is it relevant to deep learning and artificial neural networks? Well, it is important because human vision recognition and perception is one of the best designed natural neural networks, and the ultimate holy grail of deep learning is to figure out how to design deep learning networks which can match/surpass the power of human cognition. This brings us back to the question of designing efficient unsupervised deep learning networks. For instance, if we can just provide unlabelled YouTube videos to a deep neural network and allow it to learn in an unsupervised manner from the videos, what can be learnt by the neural network? For example, given a multi-layer neural network for this task, can we interpret the activity of certain nodes as recognising colour changes, certain nodes as recognising horizontal lines or certain nodes as recognising vertical lines?

The answer to the above questions is 'Yes'. There are neural networks known as auto encoders which are used for unsupervised deep learning. An auto-encoder learns the weights of the network using back propagation, wherein the expected output is set to be equal to the input. We will discuss more on auto-encoders in our next column. Also, if you were wondering about the experiment wherein an auto encoder learns without any supervision from YouTube videos, here is the paper which describes this: <https://arxiv.org/abs/1112.6209>.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, wishing all our readers a wonderful and productive month ahead! 

By: Sandya Mannarswamy

The author is an expert in systems software and is currently working as a research scientist at Xerox India Research Centre. Her interests include compilers, programming languages, file systems and natural language processing. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group 'Computer Science Interview Training India' at [http://www.linkedin.com/groups?home=HYPERLINK "http://www.linkedin.com/group s?home=&gid=2339182"&HYPERLINK "http://www.linkedin.com/groups?home=&gid=2339182"&gid=2339182](http://www.linkedin.com/groups?home=HYPERLINK%20http://www.linkedin.com/group%20s?home=&gid=2339182%20HYPERLINK%20http://www.linkedin.com/groups?home=&gid=2339182%20gid=2339182)